

Statistics:

Line Graph Reasoning

Lara the Long Distance Driver

Lara drives lorries for a popular supermarket. She has to keep track of the number of miles she drives each day in order to feed back to her manager. Here is a table to show the distance she drove on Monday.

Time	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00
Distance Travelled in Miles	0	55	115	150	205	205	260	310	365

Lara wants to construct a line graph to show her data. She decides to use the **x-axis** to show the time and the **y-axis** to show the distance travelled. Decide whether her ideas are a good idea and explain why or why not.

1. The x-axis should be labelled **Distance Travelled in km**.

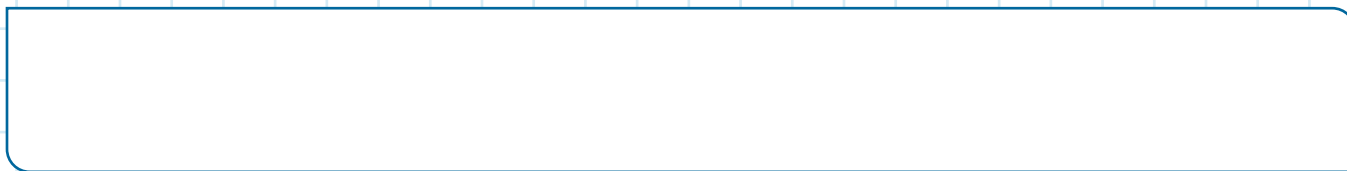
2. The scale for the y-axis should go up in increments of 10.

3. The numbers on the y-axis should start at 55 as the smallest measurement is 55 miles.

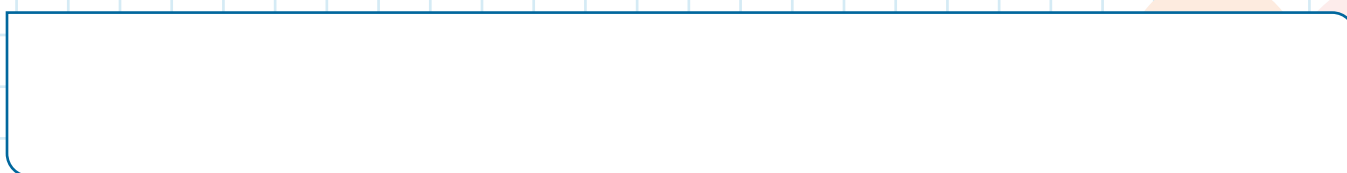
4. The y-axis should be labelled **Distance Travelled in km.**



5. The x-axis should go to 6 p.m.



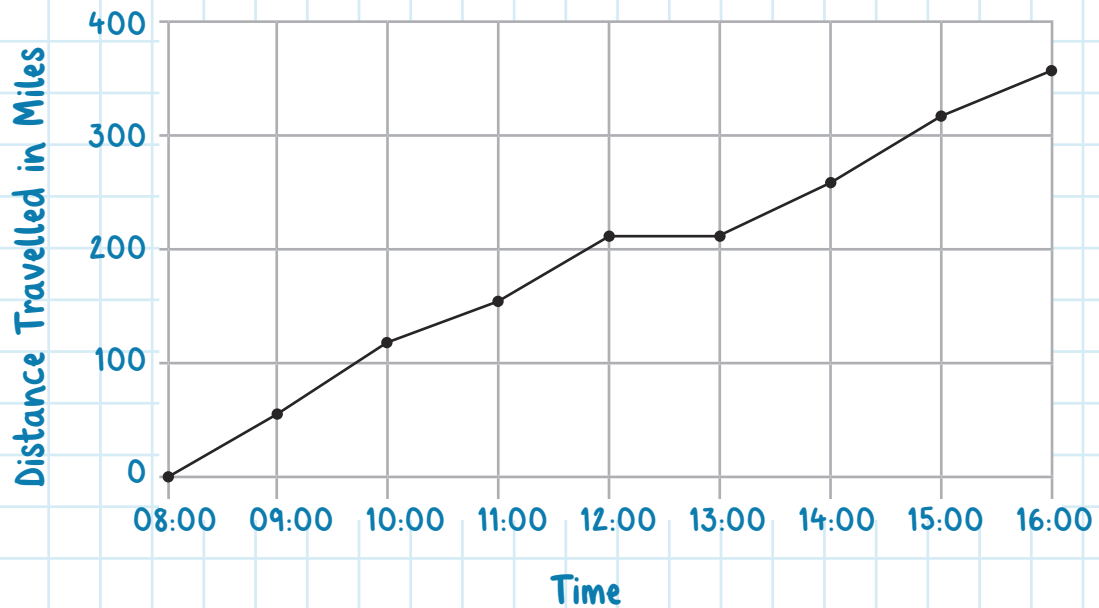
6. The time 13:00 should be missed out because no distance was travelled between 12:00 and 13:00.



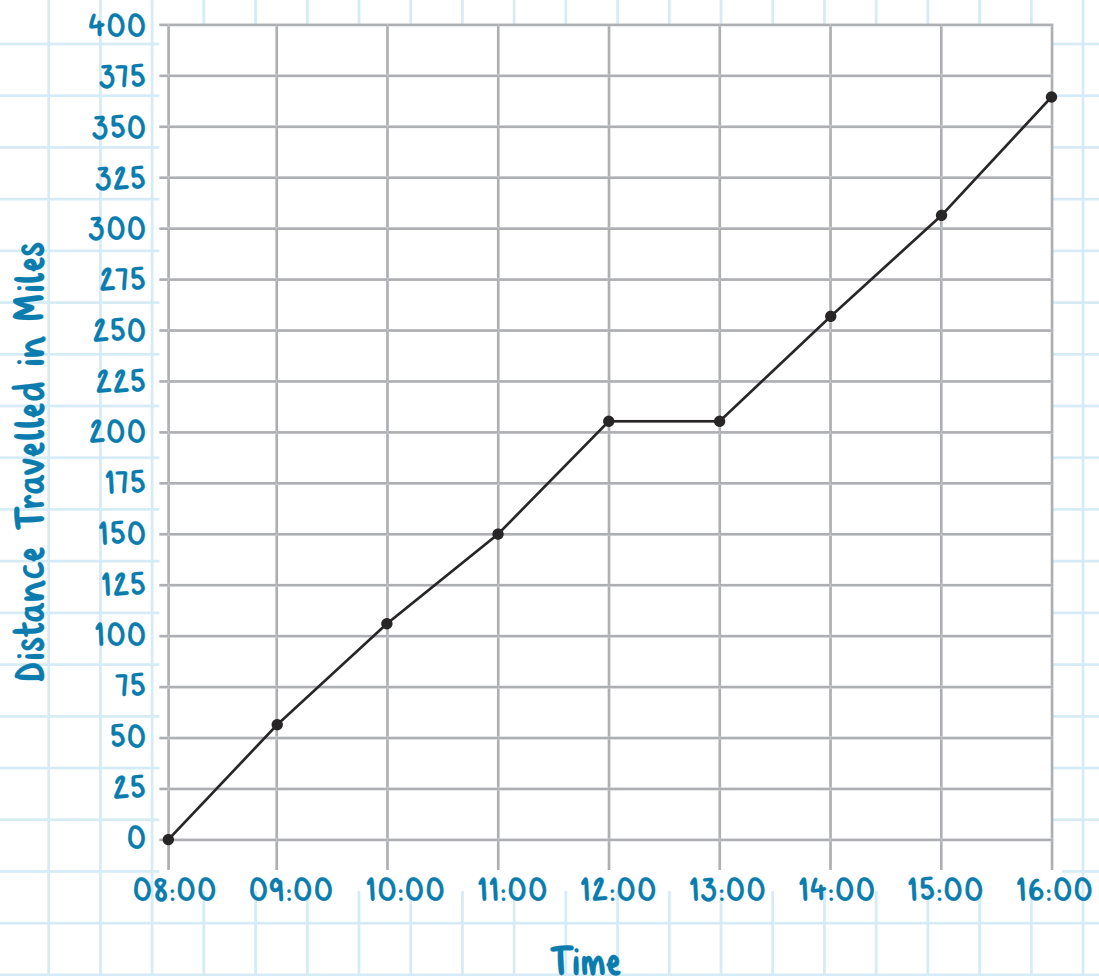
Which graph?

Lara creates two line graphs to show the data.

A Graph to Show the Distance Travelled in One Day by Lara



A Graph to Show the Distance Travelled in One Day by Lara



Which line graph do you think is the best one to use? Explain why.

Statistics: Line Graph Reasoning – Answers

Lara the Long Distance Driver

1. The x-axis should be labelled **Distance Travelled in km.**
Lara should not use this label because she is using the y-axis for the distance.
The distance travelled should also be in miles, not kilometres.
2. The scale for the y-axis should go up in increments of 10.
Lara should not use increments of 10 as she will run out of space.
The amount of increments would mean her graph is very tall.
3. The numbers on the y-axis should start at 55 as the smallest measurement is 55 miles.
She should not start the numbers at 55 as, if she does this, she would need to leave out the 08:00 data, meaning it will look as though she has not travelled at all between 08:00 and 09:00.
4. The y-axis should be labelled **Distance Travelled in km.**
Lara should not use this label because the distance travelled is being measured in miles, not kilometres.
5. The x-axis should go to 6 p.m.
The x-axis should not go to 6 p.m. because there is no data for that time. The x-axis should end at 4 p.m., which is 16:00.
6. The time 13:00 should be missed out because no distance was travelled between 12:00 and 13:00.
Lara should not leave out any data. If she left out the time 13:00, the graph would not show that she didn't travel at all during that time. This could have been her lunch time and it will be important for her manager to see that she has had a break.

Which graph?

Which line graph do you think is the best one to use? Explain why.

The second graph is a better representation of the data because it can be read more accurately to see the distances travelled by Lara at each hour. In the first line graph, 100 miles is represented by one square, which means it is much more difficult to make an accurate reading.